

ASTRO FORGE PROJECT

Technical Specifications & Simulation Parameters
Nova Foundry Team | AESS Sustainability 2026 Hackathon

1. Mass & Physical Properties

Important Note: All lengths and physical dimensions within the 3D CAD design model are specified in millimeters (mm).

The following constants define the mass distribution and inertia for the 12U CubeSat architecture:

PARAMETER	SYMBOL	VALUE	UNIT
Total Satellite Mass	m	795	kg
Moment of Inertia (X-axis)	I_{xx}	398.7	kg·m ²
Moment of Inertia (Y-axis)	I_{yy}	508.1	kg·m ²
Moment of Inertia (Z-axis)	I_{zz}	183.8	kg·m ²

2. Rotational Dynamics

The fundamental equation governing the satellite's rotational maneuverability:

$$\tau = I_{zz} \times \alpha$$

Where τ represents the applied torque and α is the angular acceleration.

3. Attitude Control System (PID Controller)

The control law used for precision targeting and satellite stabilization during debris approach:

$$u(t) = K_p e(t) + K_i \int e(\tau) d\tau + K_d (de(t)/dt)$$

Tuned Control Gains:

- Proportional Gain (K_p): **200**
- Integral Gain (K_i): **2**
- Derivative Gain (K_d): **350**

4. Eddy Current Braking System

Electromagnetic equations defining the non-contact debris braking mechanism:

$$P = I^2 R$$

$$B \approx \mu_0 (N/L) I$$

Magnetic Coil Constants:

- Number of Turns (N): **715 turns**
- Coil Resistance (R): **19.2 Ω**
- Operating Power (P): **560 W**

5. Power & Energy Storage

Energy capacity for the primary storage and the burst power delivery system:

$$E = \frac{1}{2} C V^2$$

SUBSYSTEM	CAPACITY / OUTPUT
Solar Panel Array	1000 W
Primary Storage (Li-ion Battery)	5 kWh (18 MJ)
Burst Energy (Supercapacitor)	20 kJ

6. Simulation & Thermal Configurations

MATLAB/Simulink environment parameters (V5.2 Golden Edition):

- **Coil Thermal Limit:** $T_{coil} < 150^{\circ}\text{C}$
- **Solver Configuration:** Fixed-step (ode3 Bogacki-Shampine)
- **Simulation Step Size:** 0.01 seconds (10 ms)
- **Capture Mechanism:** 6-blade Iris mechanism